

Sacha Morin

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Education

- 2022–present **PhD, Computer Science**, Université de Montréal - Mila
Machine Learning and Robotics
Expected Graduation Date: 08/2027
Advisors: Guy Wolf and Liam Paull
Labs:
○ Mila - Quebec Artificial Intelligence Institute
○ RAFALES
○ Robotics and Embodied AI Lab (REAL)
- 2021–2022 : **Research MSc, Computer Science**, Université de Montréal - Mila
Machine Learning. Not completed. Fast tracked to PhD.
Advisor: Guy Wolf
GPA: 4.30/4.30
- 2017–2021 : **Bachelor of Mathematics and Computer Science**, Université de Montréal
GPA: 4.15/4.30
- 2014–2017 : **Bachelor of Law (LLB)**, Université de Sherbrooke
GPA: 3.68/4.30

Research Experience

Mila - Quebec AI Institute

- 2022-present **Phd Student**, *Montreal, QC, Canada*
○ Robot learning and imitation learning [C1].
○ 3D multimodal scene representations for robotics and embodied AI [C2], [C3], [B1], [C4].
○ Mobile manipulation [C2].
○ Representation learning for robotics [C5], [C6].
Advisors: Guy Wolf and Liam Paull
- Apple**
- May 2026 - **ML Research Intern**, *Waterloo, Ontario, Canada*
Present **Advisors:** Alex Wong and Javad Shafiee

Samsung's SAIT AI Lab (SAIL)

March 2025 - **Research Intern**, *Montreal, QC, Canada*

- Sept 2025
- Learning from observations/videos [C1].
 - Latent action modeling.
 - Generative video models.

Advisor: Sébastien Lachapelle

Nokia Bell Labs

Summer 2024 **Research Intern**, *Murray Hill, NJ, United States*

- 3D perception with mobile robots.
- LLM integration with multimodal 3D representations.
- Train NeRFs and Gaussian splats on simulation data (Isaac Sim).

Advisor: Dan Kushnir

Mila - Quebec AI Institute

2020-2022 **Research Intern/MSc Student**, *Montreal, QC, Canada*

- Learning structured and interpretable representations by combining autoencoders and manifold learning [J4], [C7].
- Apply clustering and data visualization tools for unsupervised exploration of biological datasets [J1], [J3], [J5].

Advisor: Guy Wolf

Work Experience

LJT Laywers LLP

Jan 2020- **Lawyer (Part Time)**, *Montreal, QC, Canada*

April 2020 Mergers and acquisitions of software companies. Contract Law.

June 2019- **Articling Student**, *Montreal, QC, Canada*

Jan 2020 Mergers and acquisitions of software companies. Contract Law. Litigation.

Supervisor: Me Nicolas Lassonde

May 2018- **Law Student**, *Montreal, QC, Canada*

Aug 2018 Contract Law. Litigation.

Training

2023 **ETH Zurich Robotics Summer School**, *Avully, Switzerland*

- Search and rescue with mobile robots. Our team won the competition!

2022 **Mila Robotics Summer School**, *Montreal, QC, Canada*

- Introduction to quadruped robots and RL.

Awards

2023 **NSERC PGS D Scholarship**, Natural Sciences and Engineering Council of Canada

2023 **FRQNT Doctoral Scholarship**, Fonds de recherche du Québec - Nature et technologies

2021 **IVADO M.Sc. Scholarship**, Institut de valorisation des données

2021 **FRQNT B1X Scholarship**, Fonds de recherche du Québec - Nature et technologies

- 2021 **NSERC M.Sc. Scholarship (Declined)**, Natural Sciences and Engineering Council of Canada
- 2021 **ISM Undergraduate Research Scholarship**, Institut des sciences mathématiques
- 2020 **IVADO Undergraduate Research Scholarship**, Institut de valorisation des données
- 2019 **NSERC Undergraduate Award**, Natural Sciences and Engineering Research Council of Canada
- 2019 **Scholarship for Academic Excellence**, Bourse d'excellence des diplômés et des professeurs
- 2016 **Winner of the Matthieu-Bernard Competition**, Société québécoise de droit international
- 2015-2017 **Dean's List**, Faculty of Law, Université de Sherbrooke

Teaching and Academic Involvement

- Winter 2026 **Lecturer**, STT 3795: Theoretical Foundations of Data Science, Université de Montréal
- Taught 50% of the curriculum.
- 2023-2025 **Teaching Assistant**, STT 3795: Theoretical Foundations of Data Science, Université de Montréal
- Undergraduate class taught by Prof. Guy Wolf.
- 2023-2024 **Organizer**, Robot Learning Seminar, Mila - Quebec AI institute
- [YouTube Playlist](#)
- 2023 **Co-organizer**, Mila Robotics Summer School, Mila - Quebec AI Institute
- Prepare workshop and challenge using the Unitree Go1 robot and TagSLAM.
- 2023 **Volunteer**, Conference on Robots and Vision (CRV), Montreal, Quebec, Canada
- 2022-2023 **Member**, IVADO Student Intersectoral Committee
- Support major IVADO events, such as job fairs and *Digital October*.
- 2022 **Invited Talk on AI, Data & Algorithms**, Prof. Sylvano Santini's SEM9500 Seminar, Université du Québec à Montréal
- 2014-2015 **Pro Bono Canada**, Université de Sherbrooke
- Draft training material for directors of non-profits.

Reviewer

Journals, RA-L, T-RO

Conferences, ICML, CoRL, ICCV, RSS, IROS, ICRA, BMVC

Workshops, NeurIPS, ICLR

Software

- 2023 **StepMix**, A Python package following the scikit-learn API for model-based clustering and generalized mixture modeling of continuous and categorical data [J2].

Skills

Programming Languages Python. C++. Some knowledge of C, R, JAVA and Javascript.

Libraries PyTorch, ROS, ROS 2, Scikit-Learn, Pandas, NumPy.

Publications & Preprints

* indicates joint authorship.

Conference Proceedings

- [C1] **S. Morin**, M. Byeon, A. Jolicoeur-Martineau, and S. Lachapelle, “On the sample efficiency of inverse dynamics models for semi-supervised imitation learning”, in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, (To appear), PMLR, 2026.
- [C2] **S. Morin**, K. Gupta, M. Sandhu, C. Gauthier, F. Argenziano, K. Ellis, and L. Paull, “Agentic scene policies: Unifying space, semantics, and affordances for robot action”, in *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, (To appear), IEEE, 2026.
- [C3] C. Kassab*, **S. Morin***, M. Büchner*, M. Mattamala, K. Gupta, A. Valada, L. Paull, and M. Fallon, “Openlex3d: A new evaluation benchmark for open-vocabulary 3d scene representations”, in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025.
- [C4] Q. Gu*, A. Kuwajerwala*, **S. Morin***, K. Jatavallabhula*, B. Sen, A. Agarwal, C. Rivera, W. Paul, K. Ellis, R. Chellappa, C. Gan, C. de Melo, J. Tenenbaum, A. Torralba, F. Shkurti, and L. Paull, “Conceptgraphs: Open-vocabulary 3d scene graphs for perception and planning”, *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [C5] **S. Morin***, M. Saavedra-Ruiz*, and L. Paull, “One-4-All: Neural potential fields for embodied navigation”, in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023.
- [C6] M. Saavedra-Ruiz*, **S. Morin***, and L. Paull, “Monocular robot navigation with self-supervised pretrained vision transformers”, in *2022 19th Conference on Robots and Vision (CRV)*, IEEE, 2022, pp. 197–204.
- [C7] A. F. Duque*, **S. Morin***, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *2020 IEEE International Conference on Big Data (Big Data)*, IEEE, 2020, pp. 5027–5036.

Journal Publications

- [J1] J. S. Rhodes, A. Aumon, **S. Morin**, M. Girard, C. Larochelle, E. Brunet-Ratnasingham, A. Pagliuza, L. Marchitto, W. Zhang, A. Cutler, *et al.*, “Gaining biological insights through supervised data visualization”, *Nature Computational Science*, 2026.

- [J2] **S. Morin***, R. Legault*, F. Laliberté, Z. Bakk, C.-É. Giguère, R. de la Sablonnière, and É. Lacourse, “StepMix: A python package for pseudo-likelihood estimation of generalized mixture models with external variables”, *Journal of Statistical Software*, vol. 113, pp. 1–39, 2025.
- [J3] E. Brunet-Ratnasingham*, **S. Morin***, H. E. Randolph*, M. Labrecque, J. Bélair, R. Lima-Barbosa, A. Pagliuzza, L. Marchitto, M. Hultström, J. Niessl, *et al.*, “Sustained ifn signaling is associated with delayed development of sars-cov-2-specific immunity”, *Nature Communications*, vol. 15, no. 1, p. 4177, 2024.
- [J4] A. F. Duque*, **S. Morin***, G. Wolf, and K. R. Moon, “Geometry regularized autoencoders”, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2022.
- [J5] B. Paré, M. Rozendaal, **S. Morin**, L. Kaufmann, S. M. Simpson, R. Poujol, F. Mostefai, J.-C. Grenier, H. Xing, M. Sanchez, *et al.*, “Patient health records and whole viral genomes from an early SARS-CoV-2 outbreak in a Quebec hospital reveal features associated with favorable outcomes”, *Plos one*, vol. 16, no. 12, e0260714, 2021.

Book Chapters

- [B1] L. Paull, **S. Morin**, D. Maggio, M. Büchner, C. Cadena, A. Valada, and L. Carlone, “Towards open-world spatial ai”, in *SLAM Handbook*, To appear, 2025.

Preprints

- [P1] C. Kassab, M. Mattamala, **S. Morin**, M. Büchner, A. Valada, L. Paull, and M. Fallon, “The bare necessities: Designing simple, effective open-vocabulary scene graphs”, *arXiv preprint arXiv:2412.01539*, 2024.

Workshops

- [W1] F. Argenziano, M. Saavedra-Ruiz, **S. Morin**, D. Nardi, and L. Paull, “Dynamic objects relocalization in changing environments with flow matching”, in *Perception and Planning for Mobile Manipulation in Changing Environments @ IROS*, 2025.
- [W2] **S. Morin**, K. Gupta, M. Sandhu, C. Gauthier, F. Argenziano, K. Ellis, and L. Paull, “Agentic scene policies: Unifying space, semantics, and affordances for robot action”, in *NeurIPS 2025 Workshop on Space in Vision, Language, and Embodied AI (SpaVLE)*, 2025.
- [W3] **S. Morin***, S. Naht*, S. Ebrahimi Kahou, and G. Wolf, “Spectral temporal contrastive learning”, in *NeurIPS 2023 Workshop: Self-Supervised Learning - Theory and Practice*, 2023.
- [W4] A. F. Duque*, **S. Morin***, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *NeurIPS 2020 Workshop on Differential Geometry Meets Deep Learning (DiffGeo4DL)*, 2020.

- [W5] A. F. Duque*, **S. Morin***, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *DeepMath 2020 Conference on the Mathematical Theory of Deep Neural Networks*, 2020.
- [W6] **S. Morin***, A. F. Duque*, G. Wolf, and K. Moon, “Extendable and invertible manifold learning with geometry regularized autoencoders”, in *Montreal AI Symposium (MAIS)*, 2020.

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